

PCAP Multiple Attacks Simulation

📅 Date	August 13, 2026 → August 17, 2026
🔗 URL	https://malware-traffic-analysis.net/training-exercises.html

Attack 1 — Formbook / XLoader (C2)

Source: <https://malware-traffic-analysis.net/2026/08/09/index.html>

Password Format for unzipping: <https://www.malware-traffic-analysis.net/about.html>

This is the complete flow start to finish, including the live replay + verification:

1. Download & prep (Kali)

```
mkdir -p ~/attacks/formbook && cd ~/attacks/formbook
wget https://malware-traffic-analysis.net/2026/08/09/2026-08-09-traffic-analysis-exercise.pcap.zip
unzip -P 'infected_20260809' 2026-08-09-traffic-analysis-exercise.pcap.zip
mv 2026-08-09-traffic-analysis-exercise.pcap formbook_attack.pcap
```

```
(kali@kali)-[~]
$ mkdir -p ~/attacks/formbook && cd ~/attacks/formbook

(kali@kali)-[~/attacks/formbook]
$ wget https://malware-traffic-analysis.net/2026/08/09/2026-08-09-traffic-analysis-exercise.pcap.zip
--2026-08-13 04:00:09-- https://malware-traffic-analysis.net/2026/08/09/2026-08-09-traffic-analysis-exercise.pcap.zip
Resolving malware-traffic-analysis.net (malware-traffic-analysis.net) ... 199.201.110.204
Connecting to malware-traffic-analysis.net (malware-traffic-analysis.net)|199.201.110.204|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12792967 (12M) [application/zip]
Saving to: '2026-08-09-traffic-analysis-exercise.pcap.zip'

2026-08-09-traffic-analysis-exercise 100%[=====>] 12.20M 334KB/s
2026-08-13 04:00:54 (289 KB/s) - '2026-08-09-traffic-analysis-exercise.pcap.zip' saved [12792967/12792967]

(kali@kali)-[~/attacks/formbook]
$ unzip -P 'infected_20260809' 2026-08-09-traffic-analysis-exercise.pcap.zip
Archive: 2026-08-09-traffic-analysis-exercise.pcap.zip
  inflating: 2026-08-09-traffic-analysis-exercise.pcap

(kali@kali)-[~/attacks/formbook]
$ mv 2026-08-09-traffic-analysis-exercise.pcap formbook_attack.pcap

(kali@kali)-[~/attacks/formbook]
$ tshark -r formbook_attack.pcap -T fields -e ip.src -e ip.dst | sort -u
```

2. Identify the real local host to disguise

```
tshark -r formbook_attack.pcap -T fields -e ip.src -e ip.dst | sort -u
```

172.16.8.49 is the infected client (the C2 victim) — that's the one you swap for your real Ubuntu-Benign IP.

▼ THE FormbookC2C kali output:

```
(kali㉿kali)-[~]
└─$ mkdir -p ~/attacks/formbook && cd ~/attacks/formbook

(kali㉿kali)-[~/attacks/formbook]
└─$ wget https://malware-traffic-analysis.net/2026/08/09/2026-08-09-traffic-analysis-exercise.pcap.zip
--2026-08-13 04:00:09-- https://malware-traffic-analysis.net/2026/08/09/2026-08-09-traffic-analysis-exercise.pcap.zip
Resolving malware-traffic-analysis.net (malware-traffic-analysis.net)... 199.201.110.204
Connecting to malware-traffic-analysis.net (malware-traffic-analysis.net)|199.201.110.204|:443...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 12792967 (12M) [application/zip]
Saving to: '2026-08-09-traffic-analysis-exercise.pcap.zip'

2026-08-09-traffic-analysis-exercise 100%
[=====⇒]
12.20M 334KB/s in 43s

2026-08-13 04:00:54 (289 KB/s) - '2026-08-09-traffic-analysis-exercise.pcap.zip' saved
[12792967/12792967]

(kali㉿kali)-[~/attacks/formbook]
└─$ unzip -P 'infected_20260809' 2026-08-09-traffic-analysis-exercise.pcap.zip
Archive: 2026-08-09-traffic-analysis-exercise.pcap.zip
inflating: 2026-08-09-traffic-analysis-exercise.pcap

(kali㉿kali)-[~/attacks/formbook]
└─$ mv 2026-08-09-traffic-analysis-exercise.pcap formbook_attack.pcap

(kali㉿kali)-[~/attacks/formbook]
└─$ tshark -r formbook_attack.pcap -T fields -e ip.src -e ip.dst | sort -u

0.0.0.0 255.255.255.255
104.18.33.89 172.16.8.49
104.21.42.23 172.16.8.49
104.21.76.210 172.16.8.49
121.54.163.148 172.16.8.49
128.85.113.135 172.16.8.53
146.59.71.167 172.16.8.49
150.171.109.178 172.16.8.53
150.171.110.209 172.16.8.49
150.171.110.210 172.16.8.49
150.171.110.210 172.16.8.53
150.171.22.17 172.16.8.49
150.171.22.17 172.16.8.53
```

150.171.27.11 172.16.8.49
 150.171.27.11 172.16.8.53
 150.171.28.11 172.16.8.49
 150.171.28.11 172.16.8.53
 150.171.28.12 172.16.8.49
 150.171.28.12 172.16.8.53
 156.247.51.39 172.16.8.49
 172.16.8.49 104.18.33.89
 172.16.8.49 104.21.42.23
 172.16.8.49 104.21.76.210
 172.16.8.49 121.54.163.148
 172.16.8.49 146.59.71.167
 172.16.8.49 150.171.110.209
 172.16.8.49 150.171.110.210
 172.16.8.49 150.171.22.17
 172.16.8.49 150.171.27.11
 172.16.8.49 150.171.28.11
 172.16.8.49 150.171.28.12
 172.16.8.49 156.247.51.39
 172.16.8.49 172.16.8.255
 172.16.8.49 172.16.8.8
 172.16.8.49,172.16.8.8 172.16.8.8,172.16.8.49
 172.16.8.49 172.183.7.192
 172.16.8.49 172.183.7.194
 172.16.8.49 172.184.231.68
 172.16.8.49 172.253.113.108
 172.16.8.49 172.253.113.132
 172.16.8.49 172.64.155.76
 172.16.8.49 172.67.162.153
 172.16.8.49 172.67.219.130
 172.16.8.49 173.46.81.201
 172.16.8.49 183.90.186.205
 172.16.8.49 199.192.27.50
 172.16.8.49 199.232.210.172
 172.16.8.49 20.184.175.19
 172.16.8.49 20.184.175.4
 172.16.8.49 20.190.135.2
 172.16.8.49 20.190.155.131
 172.16.8.49 20.190.155.66
 172.16.8.49 20.42.65.88
 172.16.8.49 204.79.197.203
 172.16.8.49 20.96.236.177
 172.16.8.49 224.0.0.22
 172.16.8.49 224.0.0.251

172.16.8.49	224.0.0.252
172.16.8.49	23.103.231.14
172.16.8.49	23.200.156.146
172.16.8.49	23.200.156.216
172.16.8.49	23.220.206.14
172.16.8.49	23.2.88.44
172.16.8.49	23.33.29.197
172.16.8.49	23.33.29.206
172.16.8.49	23.33.29.213
172.16.8.49	23.38.253.107
172.16.8.49	23.47.72.95
172.16.8.49	23.66.127.39
172.16.8.49	23.66.127.53
172.16.8.49	38.182.168.246
172.16.8.49	4.150.223.108
172.16.8.49	4.150.223.110
172.16.8.49	4.150.223.97
172.16.8.49	45.130.41.161
172.16.8.49	52.110.7.23
172.16.8.49	52.123.250.140
172.16.8.49	52.168.112.67
172.16.8.49	66.29.149.91
172.16.8.49	72.145.35.111
172.16.8.49	81.2.196.19
172.16.8.49	85.210.193.152
172.16.8.49	89.110.89.25
172.16.8.53	128.85.113.135
172.16.8.53	150.171.109.178
172.16.8.53	150.171.110.210
172.16.8.53	150.171.22.17
172.16.8.53	150.171.27.11
172.16.8.53	150.171.28.11
172.16.8.53	150.171.28.12
172.16.8.53	172.16.8.255
172.16.8.53	172.16.8.8
172.16.8.53,172.16.8.8	172.16.8.8,172.16.8.53
172.16.8.53	172.183.7.192
172.16.8.53	172.183.7.193
172.16.8.53	172.64.154.167
172.16.8.53	20.184.175.4
172.16.8.53	20.190.135.5
172.16.8.53	20.190.155.132
172.16.8.53	20.190.155.3
172.16.8.53	20.190.155.67

172.16.8.53 204.79.197.203
172.16.8.53 20.96.236.177
172.16.8.53 216.198.53.3
172.16.8.53 216.198.53.6
172.16.8.53 216.198.54.6
172.16.8.53 224.0.0.22
172.16.8.53 224.0.0.251
172.16.8.53 224.0.0.252
172.16.8.53 23.200.156.143
172.16.8.53 23.200.156.216
172.16.8.53 23.2.88.44
172.16.8.53 23.33.29.201
172.16.8.53 23.33.29.203
172.16.8.53 23.33.29.207
172.16.8.53 23.33.29.211
172.16.8.53 23.33.29.213
172.16.8.53 23.33.85.235
172.16.8.53 23.66.127.43
172.16.8.53 4.150.223.97
172.16.8.53 4.207.44.77
172.16.8.53 52.123.250.155
172.16.8.53 52.168.112.67
172.16.8.53 85.210.193.152
172.16.8.8 172.16.8.255
172.16.8.8 172.16.8.49
172.16.8.8 172.16.8.53
172.16.8.8 255.255.255.255
172.183.7.192 172.16.8.49
172.183.7.192 172.16.8.53
172.183.7.193 172.16.8.53
172.183.7.194 172.16.8.49
172.184.231.68 172.16.8.49
172.253.113.108 172.16.8.49
172.253.113.132 172.16.8.49
172.64.154.167 172.16.8.53
172.64.155.76 172.16.8.49
172.67.162.153 172.16.8.49
172.67.219.130 172.16.8.49
173.46.81.201 172.16.8.49
183.90.186.205 172.16.8.49
199.192.27.50 172.16.8.49
199.232.210.172 172.16.8.49
20.184.175.19 172.16.8.49
20.184.175.4 172.16.8.49

20.184.175.4 172.16.8.53
20.190.135.2 172.16.8.49
20.190.135.5 172.16.8.53
20.190.155.131 172.16.8.49
20.190.155.132 172.16.8.53
20.190.155.3 172.16.8.53
20.190.155.66 172.16.8.49
20.190.155.67 172.16.8.53
20.42.65.88 172.16.8.49
204.79.197.203 172.16.8.49
204.79.197.203 172.16.8.53
20.96.236.177 172.16.8.49
20.96.236.177 172.16.8.53
216.198.53.3 172.16.8.53
216.198.53.6 172.16.8.53
216.198.54.6 172.16.8.53
23.103.231.14 172.16.8.49
23.200.156.143 172.16.8.53
23.200.156.146 172.16.8.49
23.200.156.216 172.16.8.49
23.200.156.216 172.16.8.53
23.220.206.14 172.16.8.49
23.2.88.44 172.16.8.49
23.2.88.44 172.16.8.53
23.33.29.197 172.16.8.49
23.33.29.201 172.16.8.53
23.33.29.203 172.16.8.53
23.33.29.206 172.16.8.49
23.33.29.207 172.16.8.53
23.33.29.211 172.16.8.53
23.33.29.213 172.16.8.49
23.33.29.213 172.16.8.53
23.33.85.235 172.16.8.53
23.38.253.107 172.16.8.49
23.47.72.95 172.16.8.49
23.66.127.39 172.16.8.49
23.66.127.43 172.16.8.53
23.66.127.53 172.16.8.49
38.182.168.246 172.16.8.49
4.150.223.108 172.16.8.49
4.150.223.110 172.16.8.49
4.150.223.97 172.16.8.49
4.150.223.97 172.16.8.53
4.207.44.77 172.16.8.53

```
45.130.41.161 172.16.8.49
52.110.7.23 172.16.8.49
52.123.250.140 172.16.8.49
52.123.250.155 172.16.8.53
52.168.112.67 172.16.8.49
52.168.112.67 172.16.8.53
66.29.149.91 172.16.8.49
72.145.35.111 172.16.8.49
81.2.196.19 172.16.8.49
85.210.193.152 172.16.8.49
85.210.193.152 172.16.8.53
89.110.89.25 172.16.8.49
```

3. Rewrite IPs using script

We are using combined script for all three now and distributing the attack:

(All the real external C2/CDN IPs — Cloudflare, etc. — are left untouched, same as before. They don't need to be reachable; the whole conversation, both directions, is already recorded in the pcap.)

Attack 2 — Web recon/exploitation probes

Source: <https://www.malware-traffic-analysis.net/2026/03/17/index.html>

This is real internet background noise — bots and scanners throwing SQLi payloads, LFI/RFI attempts, WordPress/xmlrpc probes, and webshell upload attempts at Brad's exposed web server. Unlike Formbook, there are already dozens of distinct real attacker IPs in this capture, so you don't need to fake attacker diversity — just disguise the one real destination (his server) as your Ubuntu-Benign.

1. Download & prep (Kali)

```
cd ~/attacks/webprobe
wget https://www.malware-traffic-analysis.net/2026/03/17/2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap.zip
unzip -P 'infected' 2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap.zip
mv 2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap webprobe_attack.pcap
```

```
(kali@kali)-[~]
$ mkdir -p ~/attacks/{formbook,webprobe,ftpxfil}

(kali@kali)-[~]
$ cd ~/attacks/webprobe

(kali@kali)-[~/attacks/webprobe]
$ wget https://www.malware-traffic-analysis.net/2026/03/17/2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap.zip
--2026-08-13 04:36:10-- https://www.malware-traffic-analysis.net/2026/03/17/2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap.zip
Resolving www.malware-traffic-analysis.net (www.malware-traffic-analysis.net)... 199.201.110.204
Connecting to www.malware-traffic-analysis.net (www.malware-traffic-analysis.net)|199.201.110.204|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9800439 (9.3M) [application/zip]
Saving to: '2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap.zip'

2026-03-17-seven-days-of-scans-and-p 100%[=====] 9.35M 342KB/s in 33s

2026-08-13 04:36:45 (292 KB/s) - '2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap.zip' saved [9800439/9800439]

(kali@kali)-[~/attacks/webprobe]
$ unzip -P 'infected' 2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap.zip
Archive: 2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap.zip
  skipping: 2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap  incorrect password

(kali@kali)-[~/attacks/webprobe]
$ unzip -P 'infected_20260317' 2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap.zip
Archive: 2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap.zip
  inflating: 2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap

(kali@kali)-[~/attacks/webprobe]
$ mv 2026-03-17-seven-days-of-scans-and-probes-and-web-traffic-hitting-my-web-server.pcap webprobe_attack.pcap

(kali@kali)-[~/attacks/webprobe]
$
```

2. Identify the real local host to disguise

```
(kali@kali)-[~/attacks/webprobe]
$ tshark -r webprobe_attack.pcap -Y "http" -T fields -e ip.dst | sort | uniq -c | sort -nr | head -n 10
25333 203.161.44.208
1271 195.178.110.109
1271 195.178.110.108
891 172.234.207.202
889 104.237.151.205
389 185.177.72.56
138 101.36.104.242
114 134.122.115.33
80 149.86.227.60
64 204.76.203.18

(kali@kali)-[~/attacks/webprobe]
$ tshark -r webprobe_attack.pcap -T fields -e ip.src | sort -u | wc -l
16012
```

The target web server IP in this PCAP is **203.161.44.208**.

Here is what your two outputs tell us:

1. **203.161.44.208 (25,333 HTTP packets):**

This IP received the vast majority of all HTTP traffic in the file, making it the primary victim/target server being probed. The other IPs listed (like **195.178.110.109**) received a tiny fraction of requests and are secondary targets or background noise.

2. **16,012 Unique Source IPs:**

The attack traffic originated from over 16,000 distinct IP addresses. This confirms distributed web scanning, botnet reconnaissance, or mass internet probing aimed at finding vulnerabilities on the server.

3. Rewrite IPs using script

We are using combined script for all three now and distributing the attack.

Attack 3 — GuLoader → AgentTesla with FTP data exfiltration

Source: <https://www.malware-traffic-analysis.net/2026/02/03/index.html>

Infection chain: malspam → RAR attachment → GuLoader shellcode → AgentTesla → stolen credentials exfiltrated **over FTP**, not HTTP. Since your Ubuntu-Benign already runs vsftpd, this is a nice thematic fit even though the replay doesn't require the FTP server to actually be reachable (same reasoning as before — full conversation is pre-recorded).

1. Download & prep (Kali)

```
(kali@kali)~/attacks/ftpxfil
$ wget https://www.malware-traffic-analysis.net/2026/02/03/2026-02-03-GuLoader-for-AgentTesla-style-infection-with-FTP-data-exfil.pcap.zip
--2026-08-13 04:44:08-- https://www.malware-traffic-analysis.net/2026/02/03/2026-02-03-GuLoader-for-AgentTesla-style-infection-with-FTP-data-exfil.pcap.zip
Resolving www.malware-traffic-analysis.net (www.malware-traffic-analysis.net)... 199.201.110.204
Connecting to www.malware-traffic-analysis.net (www.malware-traffic-analysis.net)|199.201.110.204|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 272898 (267K) [application/zip]
Saving to: '2026-02-03-GuLoader-for-AgentTesla-style-infection-with-FTP-data-exfil.pcap.zip'

2026-02-03-GuLoader-for-AgentTesla-s 100%[=====] 266.50K 103KB/s in 2.6s

2026-08-13 04:44:13 (103 KB/s) - '2026-02-03-GuLoader-for-AgentTesla-style-infection-with-FTP-data-exfil.pcap.zip' saved [272898/272898]

(kali@kali)~/attacks/ftpxfil
$ unzip -P 'infected_20260203' 2026-02-03-GuLoader-for-AgentTesla-style-infection-with-FTP-data-exfil.pcap.zip
Archive: 2026-02-03-GuLoader-for-AgentTesla-style-infection-with-FTP-data-exfil.pcap.zip
  inflating: 2026-02-03-GuLoader-for-AgentTesla-style-infection-with-FTP-data-exfil.pcap

(kali@kali)~/attacks/ftpxfil
$ mv "2026-02-03-GuLoader-for-AgentTesla-style-infection-with-FTP-data-exfil.pcap" ftpxfil_attack.pcap

(kali@kali)~/attacks/ftpxfil
$
```

2. Identify the real local host to disguise

```
(kali@kali)~/attacks/ftpxfil
$ tshark -r ftpxfil_attack.pcap -T fields -e ip.src -e ip.dst | sort -u
10.2.3.101      10.2.3.1
10.2.3.101      142.250.115.138
10.2.3.101      142.251.186.132
10.2.3.101      162.241.123.75
10.2.3.101      208.95.112.1
10.2.3.1        10.2.3.101
142.250.115.138 10.2.3.101
142.251.186.132 10.2.3.101
162.241.123.75  10.2.3.101
208.95.112.1    10.2.3.101
```

Why 10.2.3.101 ?

Looking at your `tshark` output, `10.2.3.101` is the common denominator in **every single connection**:

- It is talking to your gateway (`10.2.3.1`).
- It is connecting out to multiple external servers (`162.241.123.75`, `208.95.112.1`, etc.).

This identifies `10.2.3.101` as the local infected client (victim machine) performing the exfiltration/outbound traffic.

3. Rewrite IPs using script

We are using combined script for all three now and distributing the attack.

Script for rewriting IPs:

```
#!/bin/bash
# =====
====
# Multi-Attack 15 Attackers vs 5 Victims Simulation Pipeline
# Target MAC: 00:0c:29:9b:a0:64 (Suricata Interface MAC)
# =====
====

set -e

# Work & Output Directory Setup
WORKDIR="$HOME/attacks/processed_tmp"
OUTPUT_DIR="$HOME/attacks"
FINAL_PCAP="$OUTPUT_DIR/full_multi_attack_simulation.pcap"
SURICATA_MAC="00:0c:29:9b:a0:64"

# 1. Define 15 Simulated Attacker IPs
ATTACKERS=(
    10.0.0.1 10.0.0.2 10.0.0.3 10.0.0.4 10.0.0.5
    10.0.0.6 10.0.0.7 10.0.0.8 10.0.0.9 10.0.0.10
    10.0.0.11 10.0.0.12 10.0.0.13 10.0.0.14 10.0.0.15
)

# 2. Define 5 Ubuntu Victim IPs
VICTIMS=(
    192.168.93.139
    192.168.93.140
    192.168.93.141
    192.168.93.142
    192.168.93.143
)

# 3. Source Attack PCAPs
PCAPS=(
    "$HOME/attacks/formbook/formbook_attack.pcap"
    "$HOME/attacks/webprobe/webprobe_attack.pcap"
)
```

```

"$HOME/attacks/ftpxfil/ftpxfil_attack.pcap"
)

# 4. Original Victim IPs inside each corresponding PCAP
ORIG_VICTIMS=(
    "172.16.8.49"
    "203.161.44.208"
    "10.2.3.101"
)

# 5. Original Client/Attacker IPs inside each corresponding PCAP
ORIG_CLIENTS=(
    "192.168.93.130"
    "192.168.93.130"
    "192.168.93.130"
)

echo "[+] Initializing workspace at $WORKDIR..."
mkdir -p "$WORKDIR"
rm -f "$WORKDIR"/*.pcap

# -----
# Processing Loop: Rotate through PCAPs, Victims, and Attackers
# -----
i=0
for atk in "${ATTACKERS[@]"; do
    # Rotate through the 5 Victim IPs
    vic=${VICTIMS[$(( i % 5 ))]}

    # Rotate through the 3 Attack PCAPs & Original IPs
    pcap_idx=$(( i % 3 ))
    input_pcap="${PCAPS[$pcap_idx]}"
    orig_vic="${ORIG_VICTIMS[$pcap_idx]}"
    orig_cli="${ORIG_CLIENTS[$pcap_idx]}"

    if [ ! -f "$input_pcap" ]; then
        echo "[-] Error: Input file $input_pcap not found!"
        exit 1
    fi

    echo "[+] Stream $((i+1))/15: [Attacker $atk] -> [Victim $vic] using $(b
asename "$input_pcap")"

```

```

    # Rewrites packet headers while suppressing L4 soft checksum errors on truncated frames
    tcprewrite \
        --infile="$input_pcap" \
        --outfile="$WORKDIR/atk_${i}.pcap" \
        --srcipmap="$orig_cli/32:$atk/32,$orig_vic/32:$vic/32" \
        --dstipmap="$orig_cli/32:$atk/32,$orig_vic/32:$vic/32" \
        --enet-dmac="$SURICATA_MAC" \
        --fixcsum --fixhdrln \
        --skip-soft-errors 2>/dev/null

    i=$((i+1))
done

# -----
# Merge Streams into Master Simulation
# -----
echo "[+] Merging 15 processed attack streams into $FINAL_PCAP..."
mergcap -w "$FINAL_PCAP" "$WORKDIR"/atk_*.pcap

# -----
# Verification
# -----
echo "[+] Verifying IP topology in master PCAP file:"
tshark -r "$FINAL_PCAP" -T fields -e ip.src -e ip.dst | sort -u | head -n 15

echo "=====
=====
echo "[SUCCESS] Created master PCAP with 15 attackers and 5 victims at:"
echo "          $FINAL_PCAP"
echo "=====
=====

```

Successfully generated pcap with the three attacks combined and divided b/w the 15 fake attackers and 5 ubuntu victims IPs.

```
(kali@kali)-[~]
└─$ tcpdump -nn -r ~/attacks/full_multi_attack_simulation.pcap "src net 10.0.0.0/24" 2>/dev/null | awk '{print $3}' | cut -d'.' -f1-4 | awk '{count[$1]++} END {for (ip in count) print count[ip], ip}'
22387 10.0.0.1
313954 10.0.0.11
313954 10.0.0.2
353 10.0.0.12
22387 10.0.0.13
353 10.0.0.3
22387 10.0.0.4
313954 10.0.0.14
313954 10.0.0.5
353 10.0.0.15
353 10.0.0.6
22387 10.0.0.7
313954 10.0.0.8
353 10.0.0.9
22387 10.0.0.10
(kali@kali)-[~]
└─$
```

```
(kali@kali)-[~/attacks]
└─$ ~/attacks/process_multi_attack_15_vs_5.sh
[+] Initializing workspace at /home/kali/attacks/processed_tmp ...
[+] Stream 1/15: [Attacker 10.0.0.1] → [Victim 192.168.93.139] using formbook_attack.pcap
[+] Stream 2/15: [Attacker 10.0.0.2] → [Victim 192.168.93.140] using webprobe_attack.pcap
[+] Stream 3/15: [Attacker 10.0.0.3] → [Victim 192.168.93.141] using ftpexfil_attack.pcap
[+] Stream 4/15: [Attacker 10.0.0.4] → [Victim 192.168.93.142] using formbook_attack.pcap
[+] Stream 5/15: [Attacker 10.0.0.5] → [Victim 192.168.93.143] using webprobe_attack.pcap
[+] Stream 6/15: [Attacker 10.0.0.6] → [Victim 192.168.93.139] using ftpexfil_attack.pcap
[+] Stream 7/15: [Attacker 10.0.0.7] → [Victim 192.168.93.140] using formbook_attack.pcap
[+] Stream 8/15: [Attacker 10.0.0.8] → [Victim 192.168.93.141] using webprobe_attack.pcap
[+] Stream 9/15: [Attacker 10.0.0.9] → [Victim 192.168.93.142] using ftpexfil_attack.pcap
[+] Stream 10/15: [Attacker 10.0.0.10] → [Victim 192.168.93.143] using formbook_attack.pcap
[+] Stream 11/15: [Attacker 10.0.0.11] → [Victim 192.168.93.139] using webprobe_attack.pcap
[+] Stream 12/15: [Attacker 10.0.0.12] → [Victim 192.168.93.140] using ftpexfil_attack.pcap
[+] Stream 13/15: [Attacker 10.0.0.13] → [Victim 192.168.93.141] using formbook_attack.pcap
[+] Stream 14/15: [Attacker 10.0.0.14] → [Victim 192.168.93.142] using webprobe_attack.pcap
[+] Stream 15/15: [Attacker 10.0.0.15] → [Victim 192.168.93.143] using ftpexfil_attack.pcap
[+] Merging 15 processed attack streams into /home/kali/attacks/full_multi_attack_simulation.pcap ...
[+] Verifying rewritten attacker streams (10.0.0.0/24 → Targets):
10.0.0.10      192.168.93.143
10.0.0.11      192.168.93.139
10.0.0.1        192.168.93.139
10.0.0.12      192.168.93.140
10.0.0.13      192.168.93.141
10.0.0.14      192.168.93.142
10.0.0.15      192.168.93.143
10.0.0.2        192.168.93.140
10.0.0.3        192.168.93.141
10.0.0.4        192.168.93.142
10.0.0.5        192.168.93.143
10.0.0.6        192.168.93.139
10.0.0.7        192.168.93.140
10.0.0.8        192.168.93.141
10.0.0.9        192.168.93.142

[SUCCESS] Master PCAP created at: /home/kali/attacks/full_multi_attack_simulation.pcap
```

Checking in offline mode whether the suricata file generates alerts:

```

us@ubuntu:~$ mkdir -p /tmp/multi_attack_precheck
us@ubuntu:~$ sudo suricata -r ~/full_multi_attack_simulation.pcap -c /etc/suricata/suricata.yaml -l /tmp/multi_attack_precheck/ -v
[sudo: authenticate] Password:
Notice: suricata: This is Suricata version 8.0.6 RELEASE running in USER mode
Info: cpu: CPUs/cores online: 8
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Warning: detect: No rule files match the pattern /var/lib/suricata/rules/local.rules
Info: detect: 2 rule files processed. 137759 rules successfully loaded, 0 rules failed, 0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 137770 signatures processed. 17323 are IP-only rules, 5119 are inspecting packet payload, 114985 inspect application layer, 110 are decoder event only
Notice: threads: Threads created -> RX: 1 W: 8 FM: 1 FR: 1 Engine started.
Info: pcap: Starting file run for /home/us/full_multi_attack_simulation.pcap
Info: checksum: No packets with invalid checksum, assuming checksum offloading is NOT used
Info: pcap: pcap file /home/us/full_multi_attack_simulation.pcap end of file reached (pcap err code 0)
Notice: suricata: Signal Received. Stopping engine.
Info: suricata: time elapsed 73.236s
Notice: pcap: read 1 file: 1603900 packets, 318053000 bytes
Info: counters: Alerts: 99409
us@ubuntu:~$ wc -l /tmp/multi_attack_precheck/fast.log
99409 /tmp/multi_attack_precheck/fast.log
us@ubuntu:~$

```

High-Level Summary

We built an automated pipeline that took **3 base attack PCAPs** (`formbook` , `webprobe` , and `ftpxfil`) and transformed them into a single, multi-target simulation file (`full_multi_attack_simulation.pcap`) featuring **15 distinct attacker IPs** (`10.0.0.1` – `10.0.0.15`) attacking **5 victim IPs** (`192.168.93.139` – `192.168.93.143`).

Detailed Step-by-Step Breakdown

1. Topology Mapping

- **15 Simulated Attackers:** Created an array of IPs from `10.0.0.1` to `10.0.0.15` .
- **5 Target Victims:** Created an array of target IPs from `192.168.93.139` to `192.168.93.143` .
- **Round-Robin Assignment:** Modulo math (`i % 5` and `i % 3`) was used to evenly distribute the 3 attack types across the 5 victim machines across 15 separate streams.

2. Header & IP Rewriting (`tcprewrite`)

Each original attack file used completely different IP subnets in its raw capture. We mapped those specific subnets to our new simulation network:

- **Formbook Attack:** Mapped original source subnet `104.0.0.0/8` $\rightarrow$ `10.0.0.x` and original victim `172.16.8.49` $\rightarrow$ target victim IP.
- **Webprobe Attack:** Mapped original source subnet `100.0.0.0/8` $\rightarrow$ `10.0.0.x` and original victim `203.161.44.208` $\rightarrow$ target victim IP.
- **FTP Exfiltration Attack:** Mapped original source IP `10.2.3.101` $\rightarrow$ `10.0.0.x` and original destination IPs (`162.241.123.75` , `142.250.115.138` , etc.) $\rightarrow$ target victim IP.

Checksum Correction: `--fixcsum` recalculated TCP/UDP checksums so Suricata won't drop packets due to invalid checksums caused by IP modifications.

3. Layer 2 MAC Address Alignment

- Forced the Destination MAC address of every frame to `00:0c:29:9b:a0:64` using `--enet-dmac` .
- This guarantees that when you replay the traffic over the wire with `tcpreplay` , Suricata's network interface card accepts and processes the frames as intended.

4. Merging the Streams (`mergecap`)

- Combined all 15 individually modified stream PCAPs (`atk_0.pcap` through `atk_14.pcap`) into one cohesive master trace file: `/home/kali/attacks/full_multi_attack_simulation.pcap` .

5. Filtered Verification (`tshark`)

- Filtered `tshark` specifically on `ip.src == 10.0.0.0/24` to confirm that all 15 synthetic attackers were present in the final PCAP and actively sending packets to all 5 target victim IPs.

Extracted small pcap from the large one with limited packets:

```
(kali㉿kali)-[~]
$ tcpdump -r ~/attacks/full_multi_attack_simulation.pcap -c 20000 -w ~/attacks/sample_simulation.pcap
reading from file /home/kali/attacks/full_multi_attack_simulation.pcap, link-type EN10MB (Ethernet), snapshot length 65535

(kali㉿kali)-[~]
$ capinfos ~/attacks/sample_simulation.pcap
File name: /home/kali/attacks/sample_simulation.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 20 k
File size: 3,131 kB
Data size: 2,811 kB
Capture duration: 3060456.583327 seconds
Earliest packet time: 2026-02-03 11:13:50.333639
Latest packet time: 2026-03-10 22:21:26.916966
Data byte rate: 0 bytes/s
Data bit rate: 7 bits/s
Average packet size: 140.56 bytes
Average packet rate: 0 packets/s
SHA256: bf28a9f7f71e8bc240dd6e4eecd2d050f37c7ebc53ea41c7da71df33f8dd4c4c
SHA1: 375bf478fcd4206c930c621efe8f3380773cfa79
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
  Encapsulation = Ethernet (1 - ether)
  Capture length = 65535
  Time precision = microseconds (6)
  Time ticks per second = 1000000
  Number of stat entries = 0
  Number of packets = 20000

(kali㉿kali)-[~]
$
```

Live alerts are showing:

Use 12-hour clock (AM/PM)

Show reviewed / false-positive alerts

Max DB Fetch Limit: 500

Alert Filters

Leave any filter empty/blank to include everything for that field.

Severity / Confidence / Target

Severity (numeric)

Choose options

Signature Severity

Choose options

Confidence

Choose options

Attack Target

Choose options

Protocol / Category / Signature

Live Alert Stream (500 loaded in view)

Download filtered alerts as CSV

Click on any alert row below to open its full details, rule descriptions, raw JSON, and review actions.

Rows per page: 25 Page: 1 Showing page 1 of 20 (500 total rows)

SID	Formatted Time	Severity	Signature	src_ip	dest_ip	status
☒	2010937 2026-08-17 05:40:32 PM	Medium	ET SCAN Suspicious inbound to MySQL port 3306	10.0.0.11	192.168.93.139	new
☐	2010937 2026-08-17 05:40:32 PM	Medium	ET SCAN Suspicious inbound to MySQL port 3306	10.0.0.2	192.168.93.140	new
☐	2010937 2026-08-17 05:40:32 PM	Medium	ET SCAN Suspicious inbound to MySQL port 3306	10.0.0.14	192.168.93.142	new
☐	2010937 2026-08-17 05:40:32 PM	Medium	ET SCAN Suspicious inbound to MySQL port 3306	10.0.0.5	192.168.93.143	new
☐	2010937 2026-08-17 05:40:32 PM	Medium	ET SCAN Suspicious inbound to MySQL port 3306	10.0.0.8	192.168.93.141	new
☐	3400002 2026-08-17 05:40:29 PM	Medium	POSSBL PORT SCAN (NMAP -sS)	10.0.0.11	192.168.93.139	new
☐	3400002 2026-08-17 05:40:29 PM	Medium	POSSBL PORT SCAN (NMAP -sS)	10.0.0.14	192.168.93.142	new
☐	3400002 2026-08-17 05:40:29 PM	Medium	POSSBL PORT SCAN (NMAP -sS)	10.0.0.2	192.168.93.140	new
☐	3400002 2026-08-17 05:40:29 PM	Medium	POSSBL PORT SCAN (NMAP -sS)	10.0.0.5	192.168.93.143	new
☐	3400002 2026-08-17 05:40:29 PM	Medium	POSSBL PORT SCAN (NMAP -sS)	10.0.0.8	192.168.93.141	new

Detailed View for Alert ID: 2553

What we did, end to end:

1. Downloaded 3 distinct pcaps from malware-traffic-analysis.net (Formbook/XLoader C2, seven-days-of-scans web probes, GuLoader → AgentTesla FTP exfil), unzipped with each page's password, identified the real "local" host in each via `tshark`.
2. Wrote a combined `tcprewrite` script that disguised each pcap's real client/victim as one of 15 fake attacker IPs → 5 real Ubuntu-Benign IPs, forced dest MAC to the Suricata sensor's NIC, fixed checksums, and merged all 15 rewritten streams with `mergecap` into one master pcap.
3. Precheck-verified offline with `suricata -r` (99,409 alerts) — confirmed detection logic worked before touching the wire.
4. Diagnosed why the live dashboard showed nothing during actual replay: found `/var/log/suricata/eve.json` and `stats.log` had grown unbounded (369MB+/231MB) since setup, with no rotation configured.
5. Added Suricata's native `rotate-interval: day` + a `logrotate` policy (`rotate 3` , size-capped, `su` `suricata suricata` to fix a permissions error) so logs stay bounded going forward.
6. Hit a VM freeze during the first full replay attempt — root cause was the **Windows host's disk (F:) running out of space**, not the guest — freed host space, VM recovered.
7. Diagnosed a "database is locked" dashboard error — traced it to `ingestor.py` 's SQLite connection missing a `timeout / busy_timeout` (defaulting to 5s) plus a possible multi-tab race spawning duplicate ingestion threads; recommended fixes (longer timeout, a threading lock around `ingest_logs()`).
8. Diagnosed why the dashboard looked "frozen" while the tail showed live alerts — Streamlit doesn't auto-rerun on new DB rows; recommended wrapping the alert table in `@st.fragment(run_every="5s")` so it live-refreshes without manual reload.
9. Replayed the full combined pcap at a controlled `-pps=200` , watched disk/memory/eve.json live throughout — completed cleanly, no crash.
10. Confirmed on the dashboard: alerts from all 5 fake-attacker/real-victim pairs (`ET SCAN` , `SURICATA STREAM` , port-scan, SNMP/RPC signatures, etc.) are showing live, correctly attributed, sorted newest-first, with 500 rows loaded and paginating normally.